# Approximation and Rounding

Number · Edexcel IGCSE Mathematics A (4MA1) Paper 1 · Year 10 · worksheet

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| Name | Class | Date |
|---|---|---|

**Answer ALL questions.**

Write your answers in the spaces provided. You must write down all the stages in your working.

Suggested time: 60 minutes · Total marks: 60

## KEY RULES

| | |
|---|---|
| **Decimal places** | Count after the decimal point; look at the next digit: 5 or more rounds up |
| **Significant figures** | Start at the first non-zero digit; fill gaps with zeros |
| **Estimating** | Round every number to 1 significant figure, then calculate |
| **Error interval** | Use the inequality $x_{\text{min}} \leq x < x_{\text{max}}$ |

## WORKED EXAMPLE 1

[2 marks]

**Round** 4.7283 **to 3 significant figures.**

| | | |
|---|---|---|
| **Step 1** | The first significant figure is 4; count 3 figures | 4.72\|83 |
| **Step 2** | Look at the next digit: 8 is 5 or more, so round up | $4.72 \rightarrow 4.73$ |

**Answer:** 4.73

## WORKED EXAMPLE 2

[2 marks]

**Round** 0.004628 **to 2 significant figures.**

| | | |
|---|---|---|
| **Step 1** | Start at the first non-zero digit (4) | 0.0046\|28 |
| **Step 2** | The next digit is 2, less than 5, so round down | 0.0046 |

**Answer:** 0.0046

## WORKED EXAMPLE 3

[3 marks]

**Estimate the value of** $\dfrac{48.2 \times 5.1}{0.52}$

| | | |
|---|---|---|
| **Step 1** | Round each number to 1 significant figure | $\dfrac{50 \times 5}{0.5}$ |
| **Step 2** | Work out the top | $\dfrac{250}{0.5}$ |
| **Step 3** | Divide | 500 |

**Answer:** 500

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**1** Round 6.847 to 1 decimal place.

**(Total for question = 1 mark)**

*Practice question written for this worksheet (not from a past paper)*

**2** Round 0.0583 to 2 significant figures.

**(Total for question = 1 mark)**

*Practice question written for this worksheet (not from a past paper)*

**3** Round 45 719 to 3 significant figures.

**(Total for question = 1 mark)**

*Practice question written for this worksheet (not from a past paper)*

**4** Round 3.0962 to 2 decimal places.

**(Total for question = 1 mark)**

*Practice question written for this worksheet (not from a past paper)*

**5** Round 0.0004096 to 3 significant figures.

**(Total for question = 2 marks)**

*Practice question written for this worksheet (not from a past paper)*

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**6** Round 2 984 000 to 2 significant figures.

**(Total for question = 2 marks)**

*Practice question written for this worksheet (not from a past paper)*

**7** A calculator shows 7.28571428.

(a) Write this to 3 significant figures.

(b) Write this to 1 decimal place.

**(Total for question = 2 marks)**

*Practice question written for this worksheet (not from a past paper)*

**8** Estimate the value of $\frac{19.8 \times 4.1}{0.39}$

**(Total for question = 2 marks)**

*Practice question written for this worksheet (not from a past paper)*

**9** By rounding each number to one significant figure, work out an estimate for the value of

$$\frac{2.11^2 \times 58.9}{\sqrt{8.859}}$$

Show your working clearly.

.....

**(Total for question = 2 marks)**

*Edexcel IGCSE Mathematics A (4MA1) · Paper 1H · November 2025 · Question 2*

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**10** By rounding each number to one significant figure, work out an estimate for

$$31.4 \times 48.7$$

Show your working clearly.

.....

**(Total for question = 2 marks)**

*Edexcel IGCSE Mathematics A (4MA1) · Paper 1F · November 2024 · Question 16*

**11** Nav has worked out $\frac{68.3 \times 42.8}{0.021}$ on his calculator.

His answer is 139 201.9048

Without using a calculator and using suitable approximations, check that his answer is sensible.
Show your working clearly.

*Edexcel IGCSE Mathematics A (4MA1) · Paper 2HR · November 2020 · Question 6*

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**12** By rounding each number to one significant figure, work out an estimate for the value of

$$\frac{2.11^2 \times 58.9}{\sqrt{8.859}}$$

Show your working clearly.

.....

**(Total for question = 2 marks)**

*Edexcel IGCSE Mathematics A (4MA1) · Paper 1F · November 2025 · Question 19*

**13** (a) Work out the value of $\frac{2.5 + 3.6}{12.7} + \frac{8.2}{5 \times 3.6}$

Give your answer as a decimal.
Write down all the figures on your calculator display.

.....

(2)

(b) Write your answer to part (a) correct to 3 significant figures.

.....

(1)

**(Total for question = 3 marks)**

*Edexcel IGCSE Mathematics A (4MA1) · Paper 1FR · January 2023 · Question 11*

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**14** (a) Use your calculator to work out the value of $\frac{12.5}{1.4} - \sqrt{10}$

Write down all the figures on your calculator display.

.....

(2)

(b) Write your answer to part (a) correct to 2 significant figures.

.....

(1)

**(Total for question = 3 marks)**

*Edexcel IGCSE Mathematics A (4MA1) · Paper 1F · June 2026 · Question 11*

**15** (a) Use your calculator to work out the value of

$$\frac{7.45}{4.3^2 - 2.9}$$

Give your answer as a decimal.
Write down all the figures on your calculator display.

.....

(2)

(b) Write your answer to part (a) correct to 3 decimal places.

.....

(1)

**(Total for question = 3 marks)**

*Edexcel IGCSE Mathematics A (4MA1) · Paper 1F · January 2022 · Question 11*

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**16** (a) Work out the value of $\frac{\sqrt{4.6}}{8.1 - 3.7}$

Give your answer as a decimal.
Write down all the figures on your calculator display.

.....

(2)

(b) Write your answer to part (a) correct to 3 significant figures.

.....

(1)

*Edexcel IGCSE Mathematics A (4MA1) · Paper 4H · January 2016 · Question 1*

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**17** (a) Use your calculator to work out the value of

$$\frac{8.7+2.8}{1.4^2}$$

Give your answer as a decimal.
Write down all the figures on your calculator display.

.....

(2)

(b) Give your answer to part (a) correct to 2 significant figures.

.....

(1)

*Edexcel IGCSE Mathematics A (4MA1) · Paper 4H · January 2013 · Question 1*

**18** (a) Use your calculator to work out the value of

$$\frac{24.1}{8.4-7.8} - 6.2^2$$

Write down all the figures on your calculator display.

.....

(2)

(b) Give your answer to part (a) correct to 3 significant figures.

.....

(1)

*Edexcel IGCSE Mathematics A (4MA1) · Paper 3H · June 2011 · Question 1*

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**19** (a) Work out the value of $\frac{9}{12.4} + \frac{5.3 \times 2.8}{9.64}$

Give your answer as a decimal.
Write down all the figures on your calculator display.

.....

(2)

(b) Write your answer to part (a) correct to 3 significant figures.

.....

(1)

**(Total for question = 3 marks)**

*Edexcel IGCSE Mathematics A (4MA1) · Paper 1F · June 2023 · Question 14*

**20** (a) Work out the value of $\frac{\sqrt{7.4}}{9.8 - 2.1}$

Give your answer as a decimal.
Write down all the figures on your calculator display.

.....

(2)

(b) Give your answer to part (a) correct to 2 significant figures.

.....

(1)

*Edexcel IGCSE Mathematics A (4MA1) · Paper 3H · June 2013 · Question 2*

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## CHALLENGE QUESTIONS

**21** (a) Use your calculator to work out the value of

$$\frac{7.3 + 2.1}{6.4} + 2.2^2$$

Give your answer as a decimal.
Write down all the figures on your calculator display.

.....

(2)

(b) Give your answer to part (a) correct to 3 significant figures.

.....

(1)

*Edexcel IGCSE Mathematics A (4MA1) · Paper 3H · June 2017 · Question 5*

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**22** $P = a(c + y)$

$a = 8.3$ correct to 2 significant figures
$c = 2$ correct to 1 significant figure
$y = 15$ correct to the nearest 5

Work out the upper bound for the value of $P$
Show your working clearly.

*Edexcel IGCSE Mathematics A (4MA1) · Paper 2H · November 2023 · Question 17*

**23** $X = \dfrac{2a - b}{f}$

$a = 7.5$ correct to 1 decimal place.
$b = 3.42$ correct to 2 decimal places.
$f = 2$ correct to the nearest whole number

Work out the upper bound of the value of $X$
Show your working clearly.

*Edexcel IGCSE Mathematics A (4MA1) · Paper 2H · June 2022 · Question 18*

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**24**

$$G = \frac{c}{2f - 3h}$$

$c = 8$ correct to the nearest whole number
$f = 6.62$ correct to 2 decimal places
$h = 1.2$ correct to 1 decimal place

Work out the lower bound for the value of $G$
Give your answer correct to 3 decimal places.
Show your working clearly.

*Edexcel IGCSE Mathematics A (4MA1) · Paper 2H · June 2024 · Question 19*

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**25** The weight of a cake is 2.75 kg, correct to 2 decimal places.

(a) Write down the lower bound of the weight of the cake.

..... kg

(1)

Write down the upper bound of the weight of the cake.

..... kg

(1)

ıy has worked out $\frac{81.3 \times 59.2}{1.9^2}$ on her calculator.

answer is 13 332.299 17

ıy's answer is not sensible.

By rounding each number to one significant figure, work out a suitable estimate to
show that her answer is not sensible.
Show your working clearly.

(2)

**(Total for Question 5 is 4 marks)**

*Edexcel IGCSE Mathematics A (4MA1) · Paper 2H · January 2023 · Question 5*

**TOTAL FOR PAPER IS 60 MARKS**

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